The Great Moderation and the Rise of Markups

Demand Volatility and Tacit Collusion.

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July 11, 2026

Abstract

Business cycle stability has traditionally been considered beneficial for the economy. This paper identifies a cost: stable demand makes tacit collusion easier to sustain. In a repeated-oligopoly model, incentives to undercut rivals are greatest when demand peaks, so a decline in volatility increases the maximum sustainable collusive markup—most strongly in concentrated markets, with an elasticity equal to the inverse demand elasticity. The paper tests these predictions on a US state–sector panel covering the Great Moderation—the sustained decline in US macroeconomic volatility that began in the early 1980s—using a shift–share instrument for volatility based on the staggered deregulation of interstate banking. A 1% decline in volatility raises markups by 0.2%; the effect is absent in the least concentrated state–sectors and grows stronger and turns significant in the most concentrated quartile. Tacit collusion is consistent with a monopoly markup of 1.317, close to observed markups. The Great Moderation's decline in volatility accounts for up to two-thirds of the rise in markups between 1980 and 1997. More generally, the paper highlights a trade-off between macroeconomic stabilization and competition: merger guidelines should become stricter when demand stabilizes.

Keywords: markups, tacit collusion, Great Moderation, demand volatility, banking deregulation

JEL codes: D43, E32, L13, L41, G21

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The authors thank the participants at the 2023 Advances in Applied Macro Conference in Paris and the 2023 European Economic Association Conference in Barcelona.

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1 Introduction

Limiting the amplitude of the business cycle has traditionally been regarded as a key goal of macroeconomic policy. Lower output volatility reduces the welfare costs of business cycles, stabilizes employment, and anchors expectations. The policy frameworks credited with promoting it remain the template for modern stabilization policy. This paper challenges this narrative, identifying a fundamental drawback of demand stability: in concentrated markets where firms interact repeatedly, stable demand weakens firms' incentives to undercut one another, making tacit collusion easier to sustain. We develop this mechanism theoretically and test it on US state–sector data covering the Great Moderation—the sustained decline in US macroeconomic volatility that began in the early 1980s—to quantify its contribution to the rise in markups.

Our motivation is a coincidence in timing: markups began rising in the early 1980s, as documented by **?**, just as volatility fell and the Great Moderation set in—a pattern shown in Figure 1.

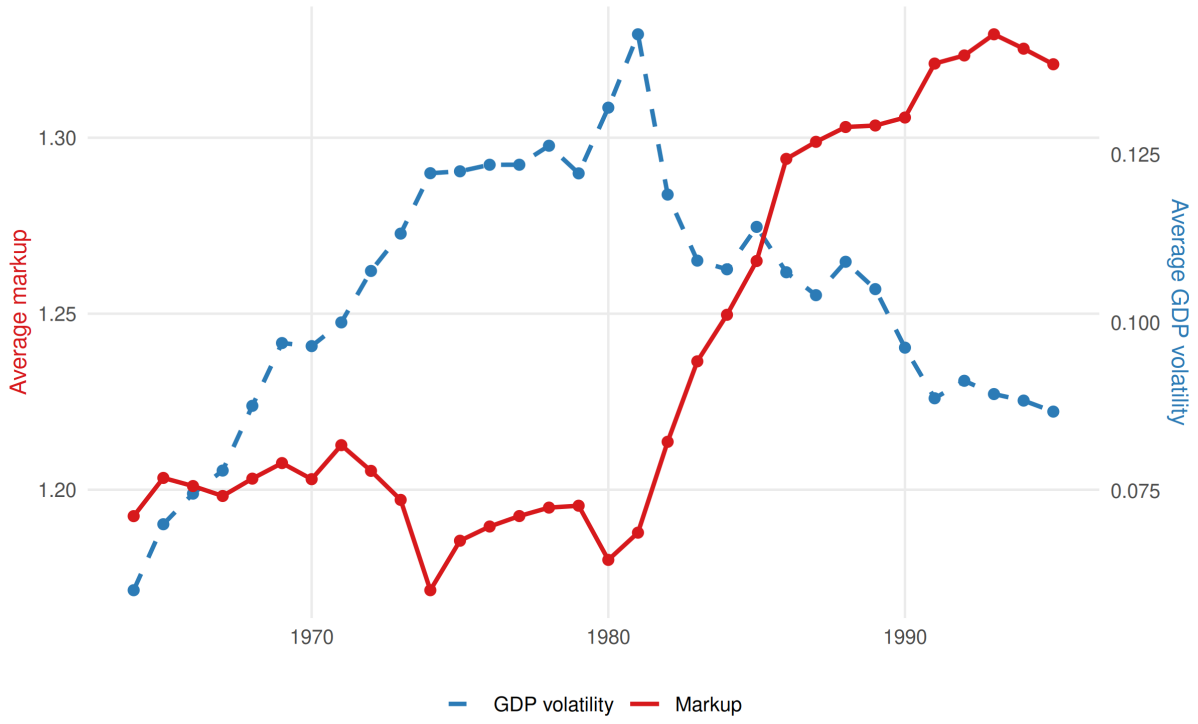


Figure 1: Sales-weighted average firm markups and sales-weighted rolling-average volatility of state–sector (SIC) GDP from 1977 to 1995, for 5-year centered rolling windows.

This markup rise is a major macroeconomic phenomenon, linked to declining business dynamism and rising inequality—two defining challenges for modern policymakers. Existing explanations are technological (the increasing importance of intangibles) or compositional (the reallocation of market shares toward high-markup superstar firms). We

are the first, to our knowledge, to provide empirical evidence for a complementary explanation, one rooted in the microeconomic theory of repeated oligopoly.

The mechanism follows from the classic analyses of tacit collusion by **?** and **?**. If firms interact repeatedly, they can sustain prices above the competitive level by threatening future retaliation. In the standard framework of **?**, the collusive equilibrium is most fragile when demand peaks and the one-shot gain from undercutting rivals is largest relative to the fixed value of future cooperation. When volatility raises the peak of the demand cycle relative to the mean, firms must therefore coordinate on a lower collusive price—or, if volatility is high enough, price at cost. When volatility falls, the cycle flattens, deviation at the peak becomes less tempting, and firms can coordinate on prices that were previously unsustainable. Section 2 formalizes this logic and derives three testable predictions: the highest sustainable markup rises as volatility falls; the channel operates only in sufficiently concentrated markets; and, when the incentive constraint binds, the elasticity of markups with respect to volatility equals the inverse of the demand elasticity.

We test these predictions on a state–sector–year panel that combines firm-level markups from Compustat with GDP data from the Bureau of Economic Analysis for 1963–1997, covering 50 states and the District of Columbia across 60 SIC-2 sectors—27,969 observations in all. We estimate markups following the production-function approach of **?** and **?**, aggregate them to the state–sector level using sales weights, and measure volatility as the backward-looking standard deviation of state–sector GDP growth. Realized volatility is endogenous to markups—market power lets firms absorb shocks through margins—and noisily measures the demand uncertainty firms actually face, so OLS estimates are plagued by reverse causality and attenuation bias. We therefore instrument volatility with the staggered deregulation of interstate banking, which every US state undertook between 1978 and 1995, interacted with predetermined baseline volatility measured over 1963–1969, before any state had deregulated. Banking integration improved risk-sharing and compressed the cross-sectional distribution of volatility: it lowered volatility in state–sectors in which baseline volatility was highest [**?**] and raised it where baseline volatility was lowest. This asymmetry supplies the differential first-stage variation. The design is a Bartik-style exposure instrument in the sense of **?**. Identification rests on the exogeneity of baseline exposure, which we test by documenting flat pre-trends and no anticipation. High- and low-exposure state–sectors do not differ systematically in predetermined markups, size, or concentration, and dropping the banking sector—where the exclusion restriction is most likely to fail—leaves the estimates essentially unchanged.

Three findings emerge. First, exogenous declines in volatility raise markups. In the headline specification with state–sector and year fixed effects, the estimated elasticity of markups with respect to volatility, evaluated at mean concentration, is -0.241: a 1% fall in volatility raises markups by roughly 0.2%. Second, as the theory predicts, the effect is largest in state–sectors that were already highly concentrated before 1978.

The marginal effect of volatility on markups is statistically insignificant in the least concentrated quartile but turns significant in higher quartiles, steepening monotonically to -0.396 in the most concentrated quartile. The estimated slopes are also quantitatively consistent with the model: under CES demand they imply a demand elasticity of 4.15 and a monopoly markup of 1.317, close to observed sales-weighted markups. The implied share of state–sectors in which the incentive constraint binds likewise rises with concentration—again as tacit collusion would predict. Third, the channel is quantitatively important: a back-of-the-envelope calculation suggests that the fall in volatility between 1980 and 1997 accounts for up to two thirds of the markup rise over the same period. These results are robust to alternative volatility windows, cost-weighted rather than sales-weighted markups, sequential sector exclusions, a leave-one-state-out jackknife, and dropping single-firm state–sectors; the volatility–HHI interaction, however, depends on including non-depository credit institutions, a sensitivity we document and discuss in Sections 5 and 7.

These findings reveal a policy trade-off between stabilizing the business cycle and preserving competition, and they show why central banks and competition agencies—national and supranational alike—need to coordinate. More concretely, when and where demand is more stable, competition authorities should scrutinize mergers more strictly, and concentration thresholds may need to tighten—especially in already concentrated markets, where repeated interaction is most likely to sustain elevated prices.

Contribution and Related Literature. The paper contributes to four literatures.

First, it contributes to the literature on the rise of market power. **?** document that the sales-weighted average markup of US publicly traded firms rose from 21% above marginal cost in 1980 to 61% in 2016, with the increase concentrated in the upper tail of the distribution. Existing explanations are largely technological or compositional: market shares reallocate toward high-markup "superstar" firms [**?**], inputs shift toward intangibles that lower marginal costs while raising fixed costs [**?**], entry costs rise and business dynamism declines [**?**], and antitrust enforcement weakens [**?**]. We add a complementary explanation based on firm conduct: as demand volatility fell, it relaxed the incentive constraint that limits tacit collusion, letting firms in concentrated markets sustain higher markups. This mechanism explains the timing of the markup rise—it begins with the Great Moderation in the early 1980s—and generates cross-sectional predictions, by concentration and by demand elasticity, that the data bear out. Our estimates imply that the channel accounts for up to two thirds of the markup rise between 1980 and 1997, a quantitatively relevant complement given the welfare costs that **?** document.

Second, the paper contributes to the literature on the Great Moderation—the decline in US macroeconomic volatility that **?** first documented and that **?** and **?** confirmed. That literature long debated whether the moderation reflected "good luck"—smaller

shocks [??]—or "good policy," particularly the post-Volcker monetary regime [??], and later emphasized structural change: ? show that the shift toward services reduced fundamental volatility while leaving the economy exposed to idiosyncratic shocks to large firms, whose granular origins ? formalizes. Throughout, this literature treats lower volatility as welfare-improving. We identify a cost of stability: by flattening the demand cycle, the moderation made tacit collusion easier to sustain, raising market power and markups.

Third, the paper puts the theory of collusion over the business cycle to work on its macroeconomic implications. Ever since ?, repeated-game models have shown that firms can sustain supracompetitive prices by threatening future retaliation. ? show that the constraint binds when demand is high, so that larger fluctuations force lower collusive prices and subsequent work extended the logic to persistent cycles, correlated shocks, and demand growth (???). ? develop the complementary imperfect-monitoring channel, under which volatility hinders the detection of deviations, and ? shows that the volatility of the discount factor—not just its level—reduces collusive prices, a mechanism we discuss in Section 7.[1] Empirically, researchers have documented collusive regimes and their breakdowns in railroads [??], retail gasoline [??], brewing [?], and vitamins [?], showing that coordination arises, softens price competition, and raises markups even absent explicit agreements. This evidence is market-specific. We are the first, to our knowledge, to provide causal evidence linking the second moment of demand to markups through the collusion channel, and to draw its aggregate implication.

Fourth, the paper draws on the literature on US interstate banking deregulation, which supplies the identifying variation. ? document its real effects on state growth, and ? show that local banking interests, rather than product-market conditions, drove the timing of deregulation—evidence that the reform is exogenous with respect to markups. Banking integration reshaped the real economy in ways later work traces out: ? show that it altered state business cycles, ? that it improved interstate income insurance, and ? that it reduced volatility most where volatility was initially highest—the asymmetry our first stage exploits. ? show that banking competition affects the market structure of non-financial sectors, precisely the direct channel that our exclusion tests in Sections 4.3 and 5 address. Methodologically, the design is a Bartik-style exposure instrument, and we follow ? and ? in placing the identifying burden on the exogeneity of baseline exposure.

The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and derives the model's testable predictions. Section 3 describes the data and measurement. Section 4 presents the empirical strategy and identifying assumptions. Section 5 reports the main results, the sectoral disaggregation, and the decomposition of the markup rise. Section 6 evaluates the compatibility of the estimates with the CES tacit-collusion benchmark. Section 7 presents robustness checks, limitations, and next

[1] ? survey the determinants of tacit collusion; ? develops the general-equilibrium implications of implicit collusion, and ? an d ? study the closely related question of markup cyclicality.

steps. Section 8 concludes.

2 Theoretical Model

In this section we develop a theoretical model of tacit collusion to illustrate the link between volatility and markups. We model the reduction of macroeconomic volatility reflected in the Great Moderation as a reduction in demand variance in a market where firms compete with an infinite horizon, allowing them to coordinate on supracompetitive prices as in **?** and **?**. The model makes two (well-known) empirical predictions: firstly, collusion becomes harder to sustain if demand variance is high; secondly, collusion is easier to sustain when there are few firms in the market.

2.1 Setup

Time is discrete and infinite, $t = 0, 1, 2, \ldots$. A single market is served by $N \geq 2$ symmetric firms producing a homogeneous good at constant marginal cost $c \geq 0$. Firms discount future payoffs at rate $\delta \in (0, 1)$. We denote the expected demand given a price $p \in \mathbb{R}$ by $D(p)$, which we assume to be decreasing in the price. We denote the monopoly price given expected demand by $p_m(c) \equiv \arg\max_p (p - c)D(p)$, which we assume exists and is unique, and is therefore weakly increasing in c by revealed preference.

Demand. In each period, market demand is

$$Q_t(p_t) = D(p_t) + \sigma\, \varepsilon_t, \tag{1}$$

where $D(\cdot)$ is twice continuously differentiable and strictly decreasing, and the shocks $\varepsilon_t \in [-1, 1]$ are i.i.d. across t with $\mathbb{E}\varepsilon_t = 0$ (mean zero) and peak realization normalized to $\max \varepsilon_t = 1$. The scalar $\sigma \geq 0$ scales the amplitude of demand fluctuations; we call it *volatility*.

We impose the following assumption.

Assumption 1. $D(p_m(c)) > \sigma$

The assumption is satisfied provided that the volatility σ and the cost c are sufficiently low that expected demand is positive. It guarantees that realized demand is non-negative for all equilibrium prices $p \leq p_m$.

Timing. We restrict attention to stationary grim-trigger equilibria in which firm can condition their prices on the current demand realization.

2.2 Sustainable Collusion

In this subsection we characterize the conditions under which collusion is sustainable. Following **?**, the collusive equilibrium is most fragile at the demand peak: the gain from undercutting rivals is largest precisely when demand—and hence the deviation profit—is highest, relative to the fixed future punishment of reversion to static Nash. The relevant incentive constraint is therefore the one evaluated in the peak state, $\varepsilon_t = 1$.

If firms coordinate on a collusive price $p > c$, each earns an equal share of expected industry profit. Because $\mathbb{E}\varepsilon_t = 0$, the per-firm continuation value is

$$V^c(p) \;=\; \mathbb{E}\left[\sum_{t=0}^{\infty} \frac{\delta^t (p-c)\big(D(p)+\sigma\varepsilon_t\big)}{N}\right] \;=\; \frac{(p-c)D(p)}{N(1-\delta)}, \tag{2}$$

which does not depend on σ (the summand is positive by Assumption 1). By contrast, volatility increases the peak demand. A firm that undercuts when demand is at its peak captures the whole market and is punished by the other firms playing the static Nash equilibrium $p = c$, resulting in $V^d = 0$ thereafter. The payoff from deviation is therefore earning $(p-c)\big(D(p)+\sigma\big)$ and the incentive constraint in the peak state is

$$\underbrace{\frac{(p-c)\big(D(p)+\sigma\big)}{N} + \delta V^c(p)}_{\text{stay collusive}} \;\geq\; \underbrace{(p-c)\big(D(p)+\sigma\big)}_{\text{deviate}}. \tag{3}$$

Substituting Equation (2) and simplifying (Appendix A) reduces (3) to a single inequality: a price $p > c$ is sustainable if and only if

$$D(p) \;\geq\; \sigma\,\chi(\delta, N), \qquad \chi(\delta, N) \;\equiv\; \frac{(N-1)(1-\delta)}{N\delta - (N-1)}, \tag{4}$$

equivalently $p \leq D^{-1}\big(\sigma\chi(\delta, N)\big)$. The scalar χ reflects the fragility of collusion and is positive if and only if $\delta > (N-1)/N$, the standard Bertrand threshold: firms must value forgone future collusive revenue above the one-shot deviation gain. Because D is strictly decreasing, Equation (4) characterizes the highest sustainable collusive price.

Lemma 1 (Highest Equilibrium Price)**.** *A collusive price $p^* > c$ exists if and only if $\delta > (N-1)/N$ and $\sigma\chi(\delta, N) < D(c)$. When these hold, the highest sustainable price $\bar{p}(\delta, \sigma, N) > c$ is the unique solution of the binding constraint*

$$D(\bar{p}) \;=\; \sigma\,\chi(\delta, N), \qquad \textit{i.e.} \qquad \bar{p} \;=\; D^{-1}\big(\sigma\chi(\delta, N)\big). \tag{5}$$

Consequently, given p_m the monopoly price given expected demand, the collusive price is

$$p^* = \min\{\bar{p}, p_m\}.$$

Proof. See Appendix A. □

Volatility enters Equation (5) only through the product $\sigma\chi$, and the comparative statics follow directly.

Proposition 1 (Volatility & the Collusive Price)**.** $\dfrac{\partial \bar{p}}{\partial \sigma} < 0.$

Proof. See Appendix A. □

To deter deviation at the peak, firms coordinate on a lower price when volatility is high, because the peak temptation grows with σ while the average collusive payoff does not. A reduction in volatility like the Great Moderation flattens the cycle and allows firms to coordinate on prices that were previously unsustainably high.

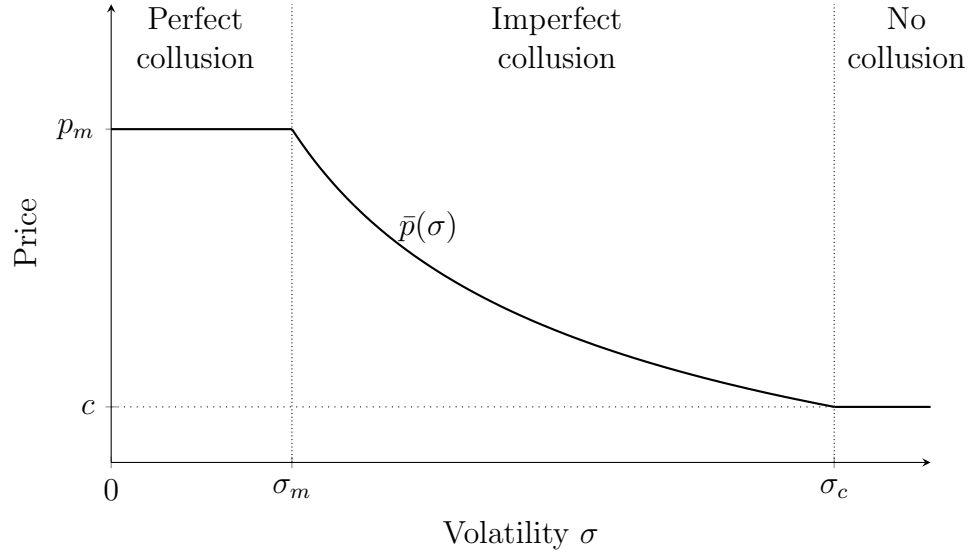


Figure 2: Sustainable collusive price.

Corollary 1 (Volatility & Markups)**.** *A decline in volatility weakly raises the highest sustainable markup, strictly if the price was below the monopoly price.*

Proposition 2 (Concentration & the Collusive Price)**.** $\dfrac{\partial \bar{p}}{\partial N} < 0.$

Proof. See Appendix A. □

With more firms, each firm's share of collusive profit shrinks while a deviation still captures the entire market, so the temptation to undercut at the peak rises and the sustainable price must fall. Under symmetric market shares, $\text{HHI} = 1/N$, so a high HHI is the empirical counterpart of a small effective number of firms. Hence, if tacit collusion is the channel through which volatility affects markups, the effect should be concentrated in high-HHI markets.

It is a well-known fact that in markets where there are few firms collusion is easier: already Adam Smith expressed his concern that "masters, being fewer in number, can combine much more easily".[2] The survey by **?** formalizes this, showing the number of firms present in the market is a key determinant of the sustainability of tacit collusion.

Corollary 2 (Concentration & Markups)**.** *For fixed δ and σ, a collusive price $\bar{p} > c$ exists only if N is small enough that $\sigma\chi(\delta, N) < D(c)$. Since χ is increasing in N, the volatility–markup channel operates only in sufficiently concentrated (high-HHI) markets.*

2.3 Markups, Volatility and Demand Elasticity

In this subsection we discuss a structural relationship between the elasticity of markups to volatility and demand elasticity that should hold if the source of the link between volatility and markups is indeed tacit collusion. Define the markup as $\mu \equiv \bar{p}/c$ and the price elasticity of average demand $\epsilon_D(p) \equiv -p\,D'(p)/D(p) > 0$. Taking logs of Equation (5), $\ln D(\bar{p}) = \ln\sigma + \ln\chi(\delta, N)$, and differentiating once with respect to $\ln\sigma$ delivers the elasticity in a single line.

Proposition 3 (Volatility Elasticity of Markups)**.**

$$\beta \;\equiv\; \frac{\partial \ln \bar{p}}{\partial \ln \sigma} \;=\; -\frac{1}{\epsilon_D(\bar{p})}. \tag{6}$$

Proof. See Appendix A. □

A 1% fall in volatility raises markups by $1/\epsilon_D$%. Hence, if the channel is tacit collusion, the Great Moderation should have raised markups most in the sectors with the most inelastic demand. Proposition 3 has a useful corollary which we will use in our calibration section.

Corollary 3 (CES Demand)**.** *The elasticity of markups with respect to volatility is constant if and only if demand is isoelastic. In particular, under CES demand with elasticity $\eta > 0$,*

$$\beta \equiv \frac{\partial \ln \mu}{\partial \ln \sigma} = -\frac{1}{\eta}. \tag{7}$$

3 Data

In this section we discuss the data used to provide evidence on the causal link between volatility and markups that we developed in the theoretical model of Section 2.

[2] *The Wealth of Nations*– Book 1 Chapter 8.

Markup Estimation. We estimate markups at the firm level, defined as the ratio of price and marginal cost. We follow the control function approach of **?** and **?**, commonly known as production function estimation. **?** popularizes this methodology and uses it to document the rise in US markups. Appendix C reports details on the data and methodology used to compute markups. The researcher has to make explicit assumptions about firms' production functions. We assume the production functions are Cobb-Douglas, following **?**. We estimate the production function for firm i in sector k at time t under the assumption of constant sector-level elasticity θ_k^v of output to variable input v.[3] Building on the Hall formula, Equation 8 shows that we can compute markups as the product of this elasticity and the ratio of output to variable input.[4] This formula follows directly from the first-order condition of firms' cost minimization:

$$\mu_{it} = \theta_k^v \times \frac{\text{Output}_{it}}{\text{Variable Input}_{it}}. \tag{8}$$

We apply this methodology to Compustat data, which covers publicly traded firms starting in 1950.[5] Although only a minority of firms operating in the US are publicly traded, they account for 41% of private sales and 29% of private employment. Moreover, collusion tends to happen between large firms, the majority of which are publicly traded.

This methodology is prone to two biases studied in detail by **?**. Firstly, we do not have information on prices or quantities. Therefore, we need to use firm revenue to measure output and hence demand elasticity biases the estimated output elasticity θ_k^v. However, this is an average-level bias, so we can still perform a regression analysis to derive meaningful results. Secondly, the control function approach proposed by **?** assumes perfect competition. Relaxing this assumption can introduce a bias in our markup estimates. **?** show that a proper first stage can account for this bias, which is small in practical applications.

We trim markups at the 1st and 99th percentile to limit the influence of extreme values. Firm-level markups are aggregated to the state–sector–year level using sales weights and cost-of-goods-sold (COGS) weights as a robustness check.

Level of Analysis. As collusion is essentially a market-level phenomenon, we need to take a stance on what constitutes a market in our analysis. We treat state–sector pairs as separate markets, reflecting the fact that demand for the same sector is typically segmented locally and that state boundaries are the most common geographic unit of analysis. This choice is noisy (in many sectors, firms operate nationwide whereas we assign them to the state where they are headquartered). However, the state–sector level is

[3] We consider 2-digit NAICS sectors.

[4] See the Appendix C and **?** and **?** for more details.

[5] This is the dataset used by **?** and **?** among others. It allows us to cover many firms and industries over a long horizon, providing reliable balance sheet data.

a useful compromise: a more aggregated definition would leave little within-year variation to exploit, while a more disaggregated one would often contain too few firms to construct stable markup measures. We therefore interpret state–sector GDP volatility as a proxy for the demand faced by firms operating in a given state–sector, and the sales-weighted state–sector markup as the empirical counterpart to $\mu \equiv \frac{p}{c}$ in the model of section 2.

GDP data. We construct volatility measures using a state–sector–year panel using GDP data from the Bureau of Economic Analysis (BEA) for the period 1963–1997. The data cover 50 states and the District of Columbia, and 60 SIC-2 industry categories. We match the BEA data to Compustat firms using the official SIC–NAICS crosswalk from the Census Bureau. As we will use the deregulation of interstate banking in U.S. states as our instrument, we limit our analysis to state–sectors that existed in 1975–77, the period immediately prior to the first state to deregulate, which was Maine in 1978. The final analysis panel contains 27,969 (27,215 with the HHI interaction) state–sector–year observations spanning 1963–1997, covering all 50 US states and the District of Columbia, up to 60 SIC sectors, and up to 1,111 state–sectors per year.

Volatility Measure. Our preferred volatility measure is the 3-year backward-looking standard deviation of log-GDP growth lagged by two periods. We treat this measure as the empirical counterpart of the theoretical volatility parameter σ in equation (1). To limit the influence of outliers, we winsorize volatility at the 99th percentile and trim at the 1st percentile.[6]

Market Concentration. We measure concentration with the Herfindahl–Hirschman index (HHI) computed from firm-level sales within each state–sector–year.[7] Denoting by $\omega_{i,sk,t} = \text{Sales}_{i,sk,t} / \sum_j \text{Sales}_{j,sk,t}$ the sales share of firm i in state–sector sk in year t, the HHI index of state–sector sk in year t is

$$\text{HHI}_{sk,t} = \sum_{i \in (sk,t)} \omega_{i,sk,t}^2 \in (0, 1], \tag{9}$$

increasing in concentration and equal to one in state–sectors with only one publicly traded firm. Our measure reflects concentration among publicly listed firms rather than concentration in the market. Moreover, there are several state–sectors with only one firm.[8] To address both issues we will run robustness checks dropping all one-firm state–

[6]Because the empirical specification uses log volatility, we trim the lower tail to avoid observations with volatility close to zero, which would generate extremely negative log values. We winsorize the upper tail rather than trim it because high-volatility state–sectors are precisely those with stronger exposure in the empirical design, and the log transformation already limits the influence of extreme high-volatility observations.

[7]See Section 4 for a discussion on the state–sector level assumption.

[8]Table 12 in Appendix D reports the number of one-firm state–sectors.

sectors.

4 Empirical Strategy

In this section we discuss our empirical strategy, detailing first the measurements of the economic objects discussed in Section 2 and then the instrumental variable regression, which allows us to claim that we identify a causal link between volatility and markups.

4.1 Measurement and Identification

Volatility Measure. The choice of backward-looking volatility is consistent with adaptive expectations rather than rational expectations, but it is easier to justify the exogeneity of interstate banking deregulation, as volatility is fully pre-determined. As the measure is backward looking the interstate banking deregulation will affect volatility slowly, possibly introducing noise in our estimates and biasing our estimates of the impact of volatility on markups towards zero. Coherently, centered and forward-looking windows yield larger coefficients and are reported as robustness checks in Section 7, together with 5-year and 7-year backward-looking windows. Our 3-year backward window stands out as the most reactive among the measures that only depend on pre-deregulation years.[9]

Baseline Volatility. As part of the instrument, we use a predetermined measure of volatility from before the deregulation period. For each state–sector, we compute the standard deviation of annual log-GDP growth using GDP data from 1963–1969, which yields growth rates for 1964–1969. State–sectors with fewer than four non-missing growth observations are excluded. We then demean this baseline volatility measure by subtracting its overall mean across state–sectors. Thus, a value of zero denotes a state–sector with average baseline volatility, while positive values denote above-average exposure. This demeaned baseline volatility is used as the exposure variable in the instrumental-variable design.

Baseline HHI. To obtain a predetermined market characteristic, we average the index over 1970–1977—the period just before the first state to deregulate, Maine in 1978—requiring at least three yearly values.[10]

[9] Figure 1 is purely descriptive and uses a 5-year centered volatility window to smooth high-frequency noise and align the plotted series with the midpoint of the volatility window.

[10] This window differs from the 1963–1969 window used for baseline volatility because the two variables play different roles. Volatility exposure enters the excluded instrument, so we measure as far from the deregulation as possible; HHI is an interaction variable, so we use the closest pre-treatment window to capture market structure immediately before deregulation.

4.2 Instrumental Variable Strategy

Why Instrument Volatility? We pursue an instrumental-variable strategy for two reasons.

First, realized volatility is endogenous to markups: firms with higher markups typically have greater pricing power and thus can absorb shocks through margins, adjusting prices rather than quantities.[11] This would lead to a spurious negative correlation between volatility and markups. Moreover, there is evidence that the rise in superstar firms made the aggregate volatility more susceptible to the idiosyncratic volatility of superstar firms.[12] This would lead to a spurious positive correlation between volatility and markups.

Second, volatility in our model is defined as a property of the distribution of demand in each period, whereas we observe only its realization. Therefore, our measure of volatility is noisy, which will imply that OLS estimators may suffer from attenuation bias. This is especially true when fixed effects are introduced and most variation is absorbed by the fixed effects.

We adopt an instrumental variable approach to address both issues, instrumenting volatility with a Bartik-style instrument with interstate banking deregulation (that occurred between 1978 and 1995) as the exogenous shifter, interacted with an exposure variable (the share variable), for which we use the average volatility in 1963–1969.

We instrument volatility of state–sector sk in period t with

$$Z_{skt} = D_{s,t-L} \times \overline{\sigma}_{sk}, \tag{10}$$

where $D_{s,t-L}$ equals one if state s had deregulated by year $t - L$ and $\overline{\sigma}_{sk}$ is the baseline volatility in 1963–69. Since our main volatility measure is a three-year backward-looking window, we set $L = 2$ so that deregulation is aligned with the beginning of the window used to construct volatility. This timing convention reduces mechanical overlap between the adoption year and the outcome window and makes the instrument predetermined relative to the measured volatility outcome.

Interstate Banking Deregulation. The Douglas Amendment to the Bank Holding Company Act of 1956 prohibited bank holding companies from establishing or acquiring bank subsidiaries across state borders unless the target bank's state authorized the transaction. Between 1978 and 1995, a wave of deregulation led 50 states and the District of Columbia to permit interstate entry. Figure 3 shows when each state legalized interstate banking: the bulk of deregulation occurred between 1982 and 1988, and most states had deregulated by 1990.

[11] See [????].
[12] See ?.

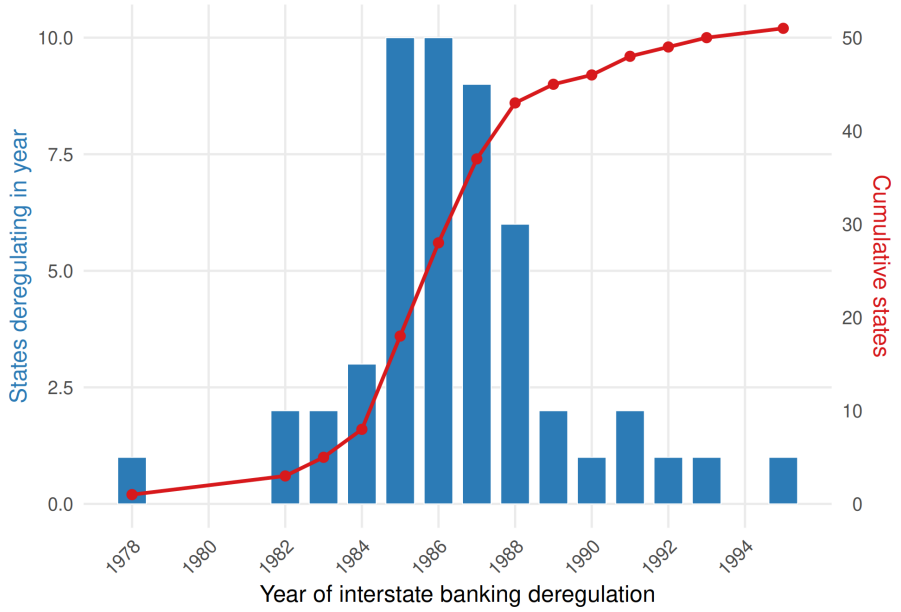


Figure 3: Staggered timing of interstate banking deregulation by state. The deregulation wave began in 1978 and ended in 1995.

? and **?** note that deregulation can raise or lower volatility depending on whether shocks are credit-supply or collateral driven; **?** finds that deregulating states experienced lower volatility through improved risk-sharing.

Baseline Volatility Exposure. Table 1 compares the volatility before and after deregulation in state–sectors with below- and above-median exposure (baseline volatility). Deregulation affected the two groups differently. In state–sectors with below-median volatility in the 1960s, mean log volatility increased significantly after deregulation. By contrast, in state–sectors with above-median baseline volatility, volatility decreased significantly. This is consistent with a risk sharing effect of the reform: by allowing banks to lend across state borders, the deregulation made state–sector cycles more similar to the aggregate business cycle, equalizing volatility across state–sectors.[13]

[13]Figure 6 in Appendix D plots the distribution of volatility before and after deregulation, separately for state–sectors with below- and above-median baseline volatility, showing that the shift affected the entire distribution.

Table 1: Deregulation, volatility and exposure

	Pre	Post	Δ (Post − Pre)
High exposure	-2.390	-2.463	-0.073***
	(0.023)	(0.020)	(0.022)
Low exposure	-3.307	-3.242	0.066**
	(0.018)	(0.024)	(0.026)
High − Low	0.917	0.779	-0.138***
	(0.028)	(0.032)	(0.037)
Observations	19,015	8,937	

Notes: state-clustered standard errors in parentheses. Groups split at the median of baseline (1963–69) exposure. The Δ column reports the within-group post-minus-pre change; the "High − Low" row the cross-sectional gap by period; the bottom-right cell is the difference-in-differences. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 1 shows a large and statistically significant differential change in volatility after deregulation by baseline exposure. Volatility falls in high-exposure state–sectors and rises in low-exposure state–sectors, consistent with deregulation equalizing volatility across state–sectors and justifying the use of baseline volatility as the exposure variable.

4.2.1 Empirical Specification

Motivated by the model's prediction that the volatility–markup channel should be stronger where collusion is easier to sustain, we let the effect of volatility vary with predetermined market concentration.

Second Stage. To test this, we let the volatility coefficient vary with predetermined concentration. Let H_{sk} denote baseline HHI, measured over 1970–77 and demeaned; we estimate, on the panel of state–sectors with Compustat coverage in 1975–1977 (predetermined relative to the earliest deregulation, Maine 1978),

$$\ln \mu_{skt} = \beta \ln \sigma_{skt} + \lambda \left(\ln \sigma_{skt} \times H_{sk} \right) + \gamma' D_{s,t-L} + \kappa \left(D_{s,t-L} \times H_{sk} \right) + \phi_{sk} + \psi_t + \epsilon_{skt}. \quad (11)$$

Here ϕ_{sk} are state–sector and ψ_t year fixed effects, absorbing time-invariant state–sector heterogeneity and common trends.[14] Standard errors are clustered by state, the level at which the shift (deregulation) varies. This allows arbitrary correlation across sectors and years within a state. Moreover, $\lambda < 0$ means the (negative) volatility effect is larger in more concentrated state–sectors, consistent with the collusion channel.

[14]Exposure and the level of H_{sk} are absorbed by the state–sector fixed effects ϕ_{sk}, so only their interaction with volatility is identified.

First Stage. The two endogenous regressors, $\ln \sigma_{skt}$ and $\ln \sigma_{skt} \times H_{sk}$, are instrumented by Z_{skt} and $Z_{skt} \times H_{sk}$, with first stages

$$e_{skt} = \alpha\, Z_{skt} + \theta \left(Z_{skt} \times H_{sk}\right) + \delta\, D_{s,t-L} + \kappa' \left(D_{s,t-L} \times H_{sk}\right) + \xi_{sk} + \zeta_t + \eta_{skt}, \qquad e_{skt} \in \{\ln \sigma_{skt},\, \ln \sigma_{skt} \times H_{sk}\}, \tag{12}$$

one for each endogenous regressor. Standard errors are again clustered by state. The marginal effect of volatility at concentration H_{sk} is

$$\frac{\partial \ln \mu_{skt}}{\partial \ln \sigma_{skt}} = \beta + \lambda\, H_{sk}, \tag{13}$$

The level of H_{sk} is absorbed by the state–sector fixed effects ϕ_{sk}, so only its interaction with volatility is identified. Because H_{sk} is demeaned, β recovers the effect at average concentration. Figure 5 plots Equation (13) across the support of H_{sk}.

We make three predictions

(i) Proposition 1 in Section 2 predicts that $\beta < 0$, as the highest collusive price is decreasing in the volatility.

(ii) Proposition 2 in Section 2 predicts that the channel is active only if the number of firms is sufficiently small, i.e. the HHI is sufficiently high. Therefore our model predicts $\lambda < 0$.

(iii) Proposition 3 in Section 2 predicts that under tacit collusion, the elasticity of markups to volatility is the inverse elasticity of demand

$$\beta + \lambda H_{sk} \approx -\frac{1}{\varepsilon_D(\bar{p})}.$$

4.3 Identifying Assumptions

Identification rests on the exogeneity of the exposure component of the instrument (baseline volatility) as discussed in **?**. Because $D_{s,t-L}$ is staggered and $\overline{\sigma}_{sk}$ is continuous, the reduced form of Z on markups is a difference-in-differences with a continuous treatment [**?**]; we make the four assumptions that design requires explicit. We report estimates based on Column (4) of Table 4, which is our headline specification.

(A1) Relevance. Conditional on the controls, the instrument Z_{skt} must shift σ_{skt}: deregulation lowers volatility more in state–sectors that were more volatile at baseline. We find evidence for relevance for both volatility and the volatility/HHI interaction.[15]

[15]The cluster-robust first-stage Wald statistics are 17.31 and 17.79, above the conventional rule of thumb of 10.

(A2) Exposure exogeneity and no anticipation. Conditional on the controls, high- and low-exposure cells would have followed parallel markup trajectories absent the volatility channel (the share exogeneity condition of **?**). We test this by regressing a dynamic reduced form of log markups on event-time × exposure, the results of which are reported in Figure 4. The closest lead is essentially zero, indicating no anticipation. The joint test over the local pre-treatment coefficients at event times -3 and -2 does not reject equality to zero. This is consistent with the absence of differential pre-trend by exposure and of no anticipation.[16]

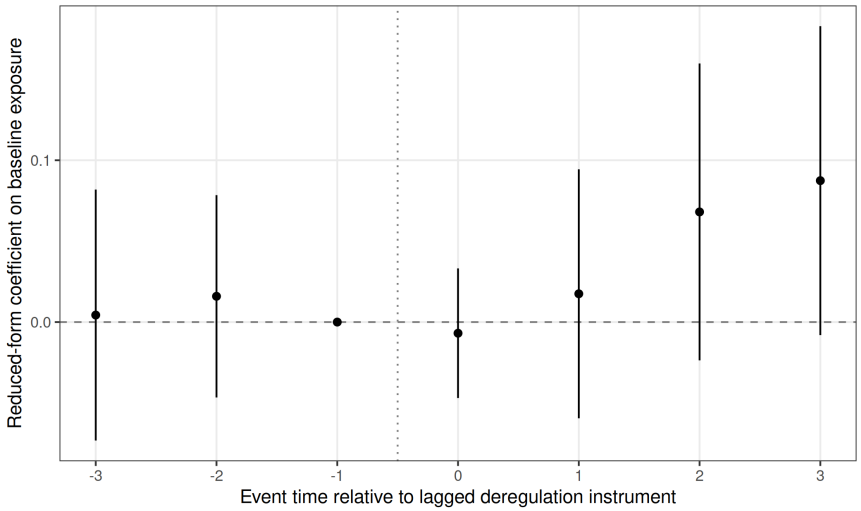


Figure 4: Dynamic Reduced Form by Baseline Exposure

Notes: The omitted category is event time -1. The specification includes state–sector and year fixed effects. Standard errors are clustered by state. Vertical bars report 95 percent confidence intervals.

Exposure exogeneity also has a cross-sectional implication. If exposure proxied for another predetermined characteristic that independently shaped markup or HHI trajectories, high- and low-exposure state–sectors should differ systematically before deregulation. Table 2 shows little evidence of this: splitting state–sectors at the median of exposure, the two groups are balanced on 1975–77 log markups, size, and concentration. The normalized differences for markups and sales are below the threshold of 0.25 specified in **?**, while that for baseline HHI is essentially at the threshold, being 0.252. This is not consistent with exposure proxying for a separate driver of markup growth. In Appendix D, Figure 8 further shows that the two groups overlap in distribution, not only in means.

[16]The dynamic reduced form regression is

$$\ln \mu_{skt} = \phi_{sk} + \psi_t + \sum_{\ell \in \mathcal{E},\, \ell \neq -1} \rho_\ell \mathbf{1}\{e_{st} = \ell\} + \sum_{\ell \in \mathcal{E},\, \ell \neq -1} \theta_\ell \left(\mathbf{1}\{e_{st} = \ell\} \times \overline{\sigma}_{sk}\right) + u_{skt}, \tag{14}$$

where $e_{st} = t - (T_s + 2)$ is event time relative to the lagged deregulation $D_{s,t-2}$, T_s is the legal deregulation year in state s, $\mathcal{E} = \{-3, -2, 0, 1, 2, 3\}$, and event time -1 is the omitted bin. The plotted coefficients are the θ_ℓ's.

Table 2: Pre-treatment balance by baseline-exposure group

	High exposure (N=485)		Low exposure (N=485)		
Variable	Mean	SD	Mean	SD	Norm. diff.
Baseline log markup	0.164	0.224	0.184	0.199	-0.093
Baseline log sales	6.155	2.403	6.702	2.355	-0.230
Baseline HHI	0.795	0.250	0.728	0.285	0.252

(A3) Exclusion. Z_{skt} must affect markups only through σ_{skt}. The exclusion restriction is not testable, so we argue it on institutional and empirical grounds. The most plausible violation is a direct effect of deregulation on markups operating through firms' credit access rather than through volatility. That channel is sharpest in banking itself, the sector the reform directly deregulated: dropping depository institutions (SIC 60) leaves the coefficient negative, significant, and essentially unchanged, suggesting that the deregulated sector did not affect results. Excluding finance (SIC 60–62) is a more conservative robustness check and, there the estimate stays negative and of comparable magnitude but the volatility/HHI interaction loses significance (see Section 5 for discussion). Moreover, the absence of pre-trends in Figure 4 is also consistent with the instrument not having a direct effect on markups.

(A4) No cross-state–sector interference (SUTVA). Deregulation in one state must not directly affect markups or volatility in other states' state–sectors. The exposure-interacted design limits this concern because identification comes from differential post-deregulation changes across state–sectors with different baseline volatility exposure, not from the average effect of deregulation itself. Spillovers common across exposure groups are absorbed by the control; only spillovers systematically correlated with exposure (baseline volatility) would threaten identification. We argue that these are second order.

5 Results

In this section we discuss our results, leading up to our headline result, Column (4) of Table 4. Volatility has a negative causal effect on markups and the effect is more negative in state–sectors with high HHI, where tacit collusion is more likely to occur. We then disaggregate the result by sector to show that the effect of volatility on markups is not driven by a specific sector, whereas the volatility/HHI interaction is driven by non-depository credit institutions (SIC 61).

5.1 Effect of Volatility on Markups

Table 3 reports OLS and IV estimates of the effect of volatility on markups. Column (1) shows a negative correlation, a 1% fall in volatility is associated with a 0.015% rise in markups. However, this relationship becomes insignificant once year and state–sector fixed effects are added in column (2), suggesting attenuation bias may be present, as discussed in Section 4.

Due to the possible attenuation bias and the reverse causality, in Columns (3) and (4) we instrument volatility using interstate banking deregulation interacted with baseline volatility in 1963–1969. The IV estimates are substantially larger in magnitude. Column (4) suggests that on average, a 1% fall in volatility increased markups by 0.2%.

The post-deregulation control is positive and significant without fixed effects, but becomes zero once state–sector and year fixed effects are included. This supports the exclusion restriction by showing that deregulation itself does not predict markups after controlling for fixed effects.

The lower panel reports the first stage. The instrument has a negative sign, coherent with **?**: deregulation caused a larger decline in volatility in state–sectors with higher baseline volatility. The first-stage is significant, and the first-stage Wald statistic is 28.56, indicating a strong first-stage relationship. The Wu–Hausman test rejects the exogeneity of volatility at the 1 percent level, supporting the use of the IV specification.

Taken together, the results suggest that exogenous reductions in volatility raise markups. This is consistent with our theoretical model: more stable demand conditions make higher collusive prices sustainable, raising markups.

5.2 Heterogeneity by Market Concentration

This is our headline result: we let the effect of volatility vary with predetermined concentration, measured by mean HHI over 1970–77. Table 4 reports it. In column (4), our preferred specification with state–sector and year fixed effects, the volatility coefficient at mean concentration is $\hat{\beta} = -0.241$ and the interaction is $\hat{\lambda} = -0.595$, both significant. The markup effect of volatility is thus close to zero in low-HHI state–sectors and more negative in high-HHI ones, matching the model's prediction that the channel operates mainly where collusion is sustainable.

The direct effects of deregulation are absent. Both the post-deregulation dummy (-0.005) and its interaction with HHI (-0.002) are small and insignificant in column (4), so deregulation has no detectable effect on markups—overall or varying with concentration—once volatility is instrumented. The HHI interaction therefore comes from the volatility channel rather than a direct concentration-specific effect of deregulation, consistent with the exclusion restriction.

The two endogenous regressors, $\ln \sigma_{skt}$ and $\ln \sigma_{skt} \times H_{sk}$, are each moved chiefly by their

Table 3: Log Volatility and Log Markups: OLS and 2SLS

Statistic	(1) OLS	(2) OLS + FE	(3) IV	(4) IV + FE
Dependent variable: log markups				
Log Volatility	-0.014***	-0.002	-0.132*	-0.207*
	(0.003)	(0.002)	(0.067)	(0.108)
Post-deregulation			0.068***	-0.004
			(0.005)	(0.010)
Exposure			0.584**	
			(0.268)	
Dependent variable: log volatility				
First stage: instrument	–	–	-1.159**	-0.843***
			(0.491)	(0.158)
First stage: post-deregulation	–	–	-0.050**	-0.084***
			(0.023)	(0.024)
First stage: exposure	–	–	4.840***	
			(0.538)	
State-sector FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
Observations	27,952	27,952	27,952	27,952
R^2	0.005	0.722	0.028	0.723
First-stage Wald			5.56	28.56
Wu-Hausman			26.25	132.55

Notes: Standard errors clustered by state in parentheses. Wald is the cluster-robust first-stage statistic. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

own instrument. In the volatility first stage the instrument enters negatively (−1.237): deregulation lowers volatility more where baseline volatility is higher, as in Table 3. Its interaction with HHI is positive (2.179), so this decline is attenuated in more concentrated markets. In the $\ln\sigma \times H$ first stage the pattern mirrors this, with the instrument–HHI interaction loading strongly and negatively (−1.590). The first-stage Wald statistics are 17.31 and 17.79, and the Wu–Hausman test rejects exogeneity at the 1 percent level, supporting the IV specification.

Table 5 evaluates the interacted marginal effect

$$\mathrm{ME}(H) \;=\; \hat{\beta} + \hat{\lambda} H_{dm} \;=\; \frac{\partial \ln \mu}{\partial \ln \sigma}$$

at each baseline-HHI quartile's mean concentration. The slope profile matches the model's two predictions: in the lowest quartile it is an insignificant near-zero ($\hat{\beta} = 0.027$), where low concentration makes coordination hard to sustain, and it steepens monotonically to −0.396 in the most concentrated quartile.

Table 5: Volatility–markup slope by baseline concentration

	Baseline HHI	$\hat{\beta}(H) = \partial \ln \mu / \partial \ln \sigma$	95% CI $(\hat{\beta}(H))$
Full sample	0.739	-0.241***	[-0.391, -0.091]
HHI Q1 (low)	0.288	0.027	[-0.138, 0.192]
HHI Q2	0.633	-0.178***	[-0.297, -0.059]
HHI Q3	0.901	-0.337***	[-0.553, -0.121]
HHI Q4 (high)	1.000	-0.396***	[-0.657, -0.135]

Notes: Stars denote two-sided tests of $H_0 : \beta(H) = 0$: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: CIs are delta-method on the joint variance of $(\hat{\beta}, \hat{\lambda})$; SE clustered by state.

Figure 5 illustrates the results by plotting the implied marginal effect of volatility on markups against baseline market concentration.

Table 4: Heterogeneity by Local Market Concentration

Statistic	(1) OLS	(2) OLS + FE	(3) IV	(4) IV + FE
Dependent variable: log markups				
Log Volatility	-0.014***	-0.002	-0.160**	-0.241***
	(0.003)	(0.002)	(0.060)	(0.077)
Log Volatility × HHI	-0.053***	-0.006	-0.277**	-0.595**
	(0.010)	(0.005)	(0.117)	(0.260)
HHI	-0.164***		-0.795**	
	(0.041)		(0.343)	
Post-deregulation	0.069***	0.009	0.065***	-0.006
	(0.005)	(0.006)	(0.005)	(0.012)
Post-deregulation × HHI	-0.010	-0.016	-0.024	-0.000
	(0.019)	(0.018)	(0.021)	(0.027)
Exposure	0.092		0.804***	
	(0.064)		(0.289)	
Dependent variable: log volatility				
First stage: instrument	–	–	-0.401	-1.185***
			(0.370)	(0.198)
First stage: instrument × HHI	–	–	-5.269**	2.327**
			(2.253)	(1.024)
First stage: post-deregulation	–	–	0.003	-0.098***
			(0.027)	(0.027)
First stage: post-deregulation × HHI	–	–	-0.352***	0.083
			(0.113)	(0.073)
First stage: HHI	–	–	0.192***	
			(0.062)	
First stage: exposure	–	–	4.775***	
			(0.560)	
Dependent variable: log volatility × HHI				
First stage: instrument	–	–	-0.684***	0.164**
			(0.143)	(0.080)
First stage: instrument × HHI	–	–	4.287***	-1.511***
			(0.617)	(0.312)
First stage: post-deregulation	–	–	-0.050***	0.009
			(0.007)	(0.010)
First stage: post-deregulation × HHI	–	–	0.254***	-0.012
			(0.045)	(0.024)
First stage: HHI	–	–	-2.967***	
			(0.023)	
First stage: exposure	–	–	0.365***	
			(0.068)	
State-sector FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
Observations	27,221	27,221	27,221	27,221
R^2	0.037	0.721	0.030	0.723
SW cond. F: log volatility			17.31	17.31
SW cond. F: log volatility × HHI			17.79	17.79
Wu-Hausman			25.58	124.90

Notes: Standard errors are clustered by state. Wald statistics are cluster-robust first-stage statistics. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. The introduction of baseline HHI reduces the observations from 27,969 to 27,215

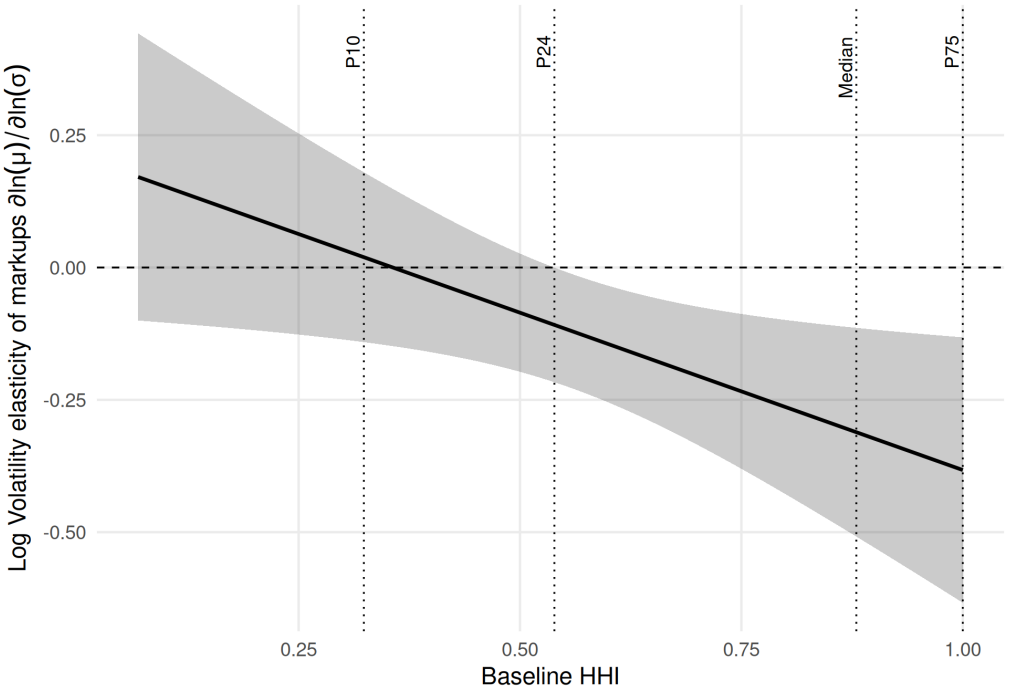


Figure 5: Marginal Effect of Log Volatility on Log Markups by HHI

Notes: Shaded bands are 95 percent confidence intervals using the delta method.

5.3 Sectoral Disaggregation

In this section we drop sectors in succession, to see whether our results are driven by a specific sector. We find that the volatility–markup relationship is robust, while the volatility/HHI interaction is driven by non-depository credit institutions.

Finance. Table 6 drops sectors where the markup measures are known to be noisy (see **?** and **?**) and finance, where we fear that the exclusion restriction might fail. Excluding agriculture, mining or real estate, or finance leaves the level coefficient at mean HHI almost unchanged and significant throughout. The volatility/HHI interaction is also stable, except when finance is removed, where it turns insignificant and shrinks in magnitude (but remains large and negative). Therefore, the negative relationship between volatility and markups, as well as the mediating role of HHI, is not driven by any specific sector. However, the financial sector contributes substantially to the precision of the interaction estimate, as the first stage relies heavily on variation from finance, reflected in the substantially lower Wald statistics when the sector is excluded. This distinction suggests that finance is important for statistical power and identification strength, but not for the economic relationship underlying our findings.

Table 6: Disaggregation: Excluding Potentially Mismeasured Sectors

Dependent Variable:	ln_mu_sale				
Specification	Baseline	No agric.	No mining	No RE	No finance
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Log Volatility	-0.241***	-0.242***	-0.203**	-0.231***	-0.241*
	(0.077)	(0.077)	(0.081)	(0.072)	(0.128)
Log Volatility × HHI	-0.595**	-0.594**	-0.501**	-0.565**	-0.532
	(0.260)	(0.260)	(0.249)	(0.249)	(0.436)
Fit statistics					
Observations	27,221	27,058	26,258	26,617	25,342
Wald (1st stage), Log Volatility	19.230	19.339	35.318	19.910	9.1918
Wald (1st stage), Log Volatility × HHI	15.511	15.672	24.058	15.632	8.5357

Notes: Agriculture (SIC 01–09, world-priced commodities), mining and oil & gas (SIC 10–14, resource rents rather than price–cost margins), real estate (SIC 65), and finance (SIC 60–62). SE clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Non-depository Credit Institutions. To investigate further the loss of precision in the interaction between volatility and concentration when we drop finance, Table 7 disaggregates the financial sector. Excluding banking (SIC 60), non-depository credit institutions (SIC 61) or securities and commodities dealer/brokers (SIC 62) leaves the level coefficient at mean HHI almost unchanged and significant throughout. The volatility/HHI interaction is also stable, except when non-depository credit institutions are removed, where it turns insignificant and shrinks (but remains large and negative). Therefore, the precision in the estimation of the interaction term hinges on non-depository credit institutions (SIC 61).

Table 7: Disaggregating the Finance Sector

Dependent Variable:	ln_mu_sale				
Specification	Baseline	No finance	No bank	No credit	No sec.
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Log Volatility	-0.241***	-0.241*	-0.260***	-0.180**	-0.276***
	(0.077)	(0.128)	(0.082)	(0.086)	(0.079)
Log Volatility × HHI	-0.595**	-0.532	-0.624**	-0.427	-0.629**
	(0.260)	(0.436)	(0.281)	(0.305)	(0.290)
Fit statistics					
Observations	27,221	25,342	26,217	26,728	26,839
Wald (1st stage), Log Volatility	19.230	9.1918	16.973	16.692	11.781
Wald (1st stage), Log Volatility × HHI	15.511	8.5357	14.349	10.332	13.640

Notes: finance denotes SIC 60–62, banking denotes depository institutions (SIC 60), credit denotes non-depository credit institutions (SIC 61), securities denotes security and commodity brokers/dealers (SIC 62). SE clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7 also provides suggestive evidence that the exclusion restriction holds: dropping banking—the sector which was directly affected by our deregulation instrument—actually makes the coefficients *larger* and more significant.

5.4 The Great Moderation and the Rise in Markups

We now ask what share of the observed markup rise the volatility channel accounts for. Comparing the start of the sample period (1980–1984) with its end (1993–1997), sales-weighted log volatility falls by $\Delta \ln \sigma^{\text{obs}} = -0.171$ while the sales-weighted log markup rises by $\Delta \ln \mu^{\text{obs}} = 0.065$.[17] Because both markups and volatility enter in logs, the concentration-interacted specification delivers an elasticity at mean concentration of $\hat{\beta} = -0.241$ (Table 4), so the markup change the channel predicts is $\hat{\beta}\,\Delta \ln \sigma^{\text{obs}}$. Attributing the entire observed decline in volatility to the Great Moderation, the implied share of the markup rise is

$$S \;=\; \frac{\hat{\beta}\,\Delta \ln \sigma^{\text{obs}}}{\Delta \ln \mu^{\text{obs}}} \;=\; 63.4\%, \qquad 95\%\ \text{CI}\ [23.8,\ 102.9]\%. \tag{15}$$

This is an upper bound: it credits the channel with the full observed stabilization, not merely the portion the instrument identifies, and it evaluates the elasticity at mean concentration. The delta-method interval treats $\Delta \ln \sigma^{\text{obs}}$ and $\Delta \ln \mu^{\text{obs}}$ as fixed and propagates only the cluster-robust variance of $\hat{\beta}$.

[17] We compare five-year averages at each end rather than single years because the volatility measure is a backward-looking window, whose level in any single early year precedes the early-1980s peak and so understates pre-moderation volatility.

6 A Benchmark for the Collusive Channel

This section uses the estimated volatility–markup elasticities as a model-based compatibility check, to check if the slopes are of a plausible magnitude under CES demand, whether they appear where the model predicts the collusive channel should operate, and what monopoly markup would be implied by the slope-estimated demand elasticity.

6.1 CES Benchmark

Under CES demand, Proposition 3 implies that, when the incentive-compatibility constraint binds and the monopoly cap is not reached, the elasticity of markups with respect to volatility satisfies

$$\beta \equiv \frac{\partial \ln \mu}{\partial \ln \sigma} = -\frac{1}{\eta}. \tag{16}$$

If the collusive channel is inactive, or if firms already reach the monopoly price, the markup is locally insensitive to volatility and $\beta = 0$. Thus the model implies the weaker restriction

$$-\frac{1}{\eta} \leq \beta \leq 0. \tag{17}$$

When the channel is active, the slope can be inverted to obtain the demand elasticity implied by the volatility response:

$$\hat{\eta}_{\text{slope}} = -\frac{1}{\hat{\beta}}.$$

This inversion is meaningful only for groups with a negative estimated elasticity. When $\hat{\beta}$ is statistically indistinguishable from zero, either tacit collusion is not sustainable or the collusive price is already the monopoly price.

Under CES demand, the monopoly markup associated with elasticity $\eta > 1$ is

$$\mu^{\star}(\eta) = \frac{\eta}{\eta - 1}.$$

Substituting the slope-implied elasticity gives the monopoly markup implied by the estimated volatility response,

$$\mu^{\star}_{\text{slope}} = \frac{\hat{\eta}_{\text{slope}}}{\hat{\eta}_{\text{slope}} - 1} = \frac{1}{1 + \hat{\beta}}.$$

We compare this benchmark with the realized sales-weighted markup $\hat{\mu}$ through the wedge

$$\Delta \equiv \mu^{\star}_{\text{slope}} - \hat{\mu}.$$

Each row of Table 5 reports a single slope $\hat{\beta}(H)$ for each HHI quartile, but the state–sectors within a quartile need not share the same collusive regime. By Corollary 3, a state–

sector whose incentive constraint binds in the interior (the middle region of Figure 2) has slope $-1/\eta$; the markup charged by firms in state–sectors whose price is at the monopoly cap or competitive is locally insensitive to volatility, so the state–sector's elasticity of markups to volatility is 0. The quartile coefficient averages $-1/\eta$ over the binding state–sectors with 0 over the rest, and so lies in $[-1/\eta, 0]$. This makes the CES inversion conservative.

Proposition 4 (CES inversion)**.** *Under CES demand with elasticity $\eta > 1$, suppose that within an HHI quartile the volatility–markup slope equals $-1/\eta$ on a share of state–sectors $q \in [0,1]$ and 0 on a share $1-q$, so that $\hat{\beta} = -q/\eta$. Then*

(i) $\hat{\beta} \in [-1/\eta,\, 0]$*, hence* $\hat{\eta}_{\text{slope}} \equiv -1/\hat{\beta} = \eta/q \;\geq\; \eta$*;*

(ii) $\mu^{\star}_{\text{slope}} = \dfrac{1}{1+\hat{\beta}} = \dfrac{\eta}{\eta - q} \;\leq\; \dfrac{\eta}{\eta-1} = \mu^{\star}(\eta)$*, with equality iff $q = 1$;*

(iii) consequently $\Delta \geq 0$ implies $\hat{\mu} \leq \mu^{\star}(\eta)$ with slack whenever $q < 1$.

Proof. See Appendix A. □

The benchmark $\mu^{\star}_{\text{slope}}$ is thus a lower bound on the true monopoly markup $\mu^{\star}(\eta)$, tightening as $q \to 1$. A positive wedge $\Delta \geq 0$ is evidence for the collusive channel; a small negative wedge is not evidence against it, since the benchmark understates $\mu^{\star}(\eta)$.

Remark 1. *The bound is exact under CES, where the only gap between $\hat{\eta}_{\text{slope}}$ and η is the censoring at the monopoly cap. Without CES, $\hat{\beta}$ recovers $-1/\epsilon_D(\bar{p})$ rather than $-1/\epsilon_D(p_m)$ (Proposition 3); since $\bar{p} \leq p_m$ a second gap opens whose sign depends on whether ϵ_D rises or falls in price.*

6.2 Concentration Profile

Now we use the HHI-quartile volatility elasticities reported in Table 5 to recover the implied demand elasticity, $\hat{\eta}_{\text{slope}} = -1/\hat{\beta}$, the implied monopoly markup, $\mu^{\star} = 1/(1+\hat{\beta})$, and the implied share of state–sectors for which the incentive-compatibility constraint binds.

Table 8 reports the results. The final column translates each slope into the share of markets in which collusion binds. By Proposition 4, $\hat{\beta}(H)$ averages a slope of $-1/\eta$ over the binding markets with zero elsewhere, so $\hat{q}(H) = \hat{\beta}(H)/\hat{\beta}(H{=}1)$ recovers the share.

In the full sample, the slope implies $\hat{\eta}_{\text{slope}} = 4.15$ and the conservative monopoly-markup benchmark $\mu^{\star} = 1.317$. This exceeds the observed markup $\hat{\mu} = 1.242$, producing a positive wedge $\Delta = 0.075$. By Proposition 4, a positive wedge is sufficient for compatibility with the CES collusive benchmark, whereas a small negative wedge would not by itself reject it.

As we consider increasing quartiles of HHI, we see that the implied elasticity of demand falls from 6.04 to 2.88 and the monopoly markup rises from 1.198 to 1.532. The impact of reducing volatility until the IC constraint stops binding Δ is increasing, coherent with the increasing share q of state–sectors which are implied to feature tacit collusion, which rises from 0.476 to 1. Intuitively, as the IC constraint is binding for more state–sectors, the impact of a reduction of volatility which would make it not bind is larger for state–sectors in the highest HHI quartile.

Table 8: Implied CES Demand Elasticity and Share of Collusive State–Sectors

	Baseline HHI	$\hat{\eta}_{\text{slope}} = -1/\hat{\beta}$	$\mu^\star = \frac{1}{1+\hat{\beta}}$	$\hat{\mu}$ (data)	$\Delta = \mu^\star - \hat{\mu}$	$\hat{q}$
Full sample	0.739	4.15	1.317	1.242	0.075	0.608
HHI Q1 (low)	0.288	–	–	1.234	–	0.000†
HHI Q2	0.633	5.63	1.216	1.262	-0.046	0.449
HHI Q3	0.901	2.97	1.509	1.274	0.235	0.851
HHI Q4 (high)	1.000	2.52	1.656	1.175	0.481	1.000

Notes: † indicates that the underlying slope $\hat{\beta}(H)$ is not significantly negative at the 5% level.

Notes. Entries are "–" where the slope is not significantly negative. CIs are delta-method holding $\hat{\mu}$ and the normalization node fixed; SE clustered by state.

In the second quartile the negative wedge suggests that the estimated slope is attenuated by the fact that only a subset of state–sectors is predicted to be in the binding-collusion region. The resulting $\mu^\star_{\text{slope}}$ understates the true CES monopoly markup, so a small negative value of Δ can arise even when realized markups remain below the relevant monopoly cap.

7 Robustness Checks

In this section we perform several robustness checks, to compare alternative volatility measures and to check whether our results are driven by specific sectors. More robustness checks are in Appendix E.

Alternative Volatility Measures. Table 9 re-estimates Column (4) of Table 4 using alternative volatility windows. Starting from our preferred three-year backward-looking measure, we consider centered and forward-looking three-year windows, as well as five- and seven-year backward-looking windows. The volatility coefficient at mean concentration and the interaction with baseline HHI remain negative across all specifications, and both are statistically significant in every column.

The pattern is informative. Moving from the backward-looking to the centered and forward-looking three-year windows increases the magnitude of both coefficients. This is consistent with the backward-looking measure being a conservative proxy for the volatility

environment relevant to firms, since centered and forward-looking measures incorporate more contemporaneous or future information but are less clean from a causal-timing perspective. By contrast, extending the backward-looking window from three to five and seven years attenuates the estimated coefficients, while strengthening the first stage. A natural interpretation is that longer windows smooth transitory variation in volatility and are more strongly predicted by baseline exposure, but they also dilute the timing of the markup response. Overall, the negative volatility–markup relationship and its amplification by concentration do not depend on the particular volatility window.

Table 9: Alternative Volatility Windows and Directions

Dependent Variable:	ln_mu_sale				
Specification	Backward 3y	Centered 3y	Forward 3y	Backward 5y	Backward 7y
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Log Volatility	-0.241***	-0.308***	-0.336***	-0.192***	-0.150***
	(0.077)	(0.094)	(0.120)	(0.069)	(0.056)
Log Volatility × HHI	-0.595**	-0.783**	-0.860*	-0.503**	-0.373*
	(0.260)	(0.331)	(0.430)	(0.239)	(0.195)
Fit statistics					
Observations	27,221	27,517	27,690	26,508	25,543
Wald (1st stage), Log Volatility	19.230	14.814	15.086	34.816	48.121
Wald (1st stage), Log Volatility × HHI	15.511	11.701	12.055	25.650	31.052

Notes: the instrument is dated to the window midpoint ($L = 2, 3, 4$ for the 3-, 5-, 7-year backward windows; $L = 0$ for centered and forward). SE clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Supplementary Checks. Appendix E reports several robustness checks. Table 14 presents the regional disaggregation, while Table 15 implements a jackknife exercise that drops large states. Tables 16 and 17 further exclude all state–sectors with only one firm. The headline results are unaffected: the significance of the volatility/HHI interaction coefficient depends on finance, and more specifically on the sector of non-depository credit institutions.

7.1 Limitations and Next Steps

Potential financing-channel endogeneity. A remaining concern is that interstate banking deregulation may affect measured markups through a financing channel rather than through product-market power. In particular, deregulation may relax credit constraints for capital-intensive non-financial firms, allowing them to scale up and mechanically increase variable-cost markups, $\theta^v \cdot \text{sales/COGS}$, through changes in cost structure. This channel is distinct from within-finance repricing, which is addressed by excluding

financial sectors, SIC 60–62.

A natural next step is to test this mechanism directly using a predetermined, demeaned measure of financial dependence, F_{sk}. This measure could be constructed either from the external-finance dependence index of **?** or from baseline PPE-to-assets. The key test would ask whether the estimated effect of deregulation-induced volatility changes is systematically stronger in financially dependent state–sectors.

Event-study evidence on deregulation. The empirical design would also benefit from event-study evidence on the effect of staggered interstate banking deregulation on volatility and markups. Such evidence would strengthen the exclusion restriction by showing whether volatility responds at the expected time and whether markups adjust only after deregulation. It would also be independently useful, since the effect of interstate banking deregulation on product-market markups has not, to our knowledge, been directly studied in the literature.

Real interest rates. The Great Moderation coincided not only with a decline in volatility, but also with a decline in the real interest rates faced by firms, which had a downward trend from 1981 to 2008. Under the standard risk-neutral benchmark, the real interest rate pins down the discount factor through

$$\delta = \frac{1}{1+r}.$$

A lower real interest rate therefore raises δ, making future collusive profits more valuable relative to current deviation profits. In repeated-game models of tacit collusion, this makes collusion easier to sustain. This provides an alternative mechanism through which the Great Moderation may have raised markups: not by reducing volatility directly, but by lowering the cost of transferring future profits into the present. Our empirical design partially addresses this concern. Common aggregate movements in real interest rates are absorbed by year fixed effects, while time-invariant differences in state–sector borrowing conditions are absorbed by state–sector fixed effects.

Monitoring and imperfect information. A related interpretation is based on imperfect monitoring. In **?**, collusion is harder to sustain when firms cannot perfectly distinguish between adverse market conditions and secret deviations. If volatility makes observed outcomes a noisier signal of firms' conduct, then high volatility weakens the ability to detect and punish deviations. Under this interpretation, the Great Moderation may have facilitated collusion by making deviations easier to identify and punishment strategies more effective. This monitoring channel is distinct from our model, which is based on **?**, and closer to the logic of **?** and to evidence based on firm-level uncertainty

and idiosyncratic markup volatility.

8 Conclusion

Greater stability of the demand cycle has traditionally been viewed as unambiguously beneficial to the economy. This paper challenges that view by documenting a significant drawback of greater stability. In repeated oligopoly, tacit collusion is most fragile when demand peaks and the temptation to undercut rivals is strongest; when volatility falls, the cycle flattens, that temptation weakens, and collusive prices that were previously unsustainable become sustainable. We formalize this mechanism in the framework of **?** and test its predictions on a US state–sector panel spanning the Great Moderation—the reduction of US business cycle volatility starting in 1980—instrumenting volatility with the staggered deregulation of interstate banking interacted with predetermined baseline volatility. The evidence is consistent with the theory in three respects. First, exogenous declines in volatility raise markups, with an elasticity of -0.241 at mean state–sector concentration. Second, the effect is absent in the least concentrated state–sectors and steepens monotonically to -0.396 in the most concentrated quartile—precisely where repeated interaction can sustain coordination. Third, the estimated slopes are quantitatively consistent with the CES tacit-collusion benchmark: they imply a demand elasticity of 4.15 and a monopoly markup of 1.317, close to observed sales-weighted markups, with the implied share of collusive state–sectors rising with concentration. A back-of-the-envelope calculation attributes up to two thirds of the 1980–1997 markup rise to the volatility channel. The rise in markups, usually explained by technology or composition, thus also reflects a change in competitive conduct: macroeconomic stability made tacit collusion easier to sustain. The research design identifies a causal effect of volatility on markups. The collusive reading of that effect rests on the concentration heterogeneity, the sectoral patterns, and the quantitative fit with the CES benchmark, all of which align with the repeated-interaction mechanism but do not exhaust the alternative explanations; in particular, the volatility–concentration interaction depends on including non-depository credit institutions, a sensitivity we document in Sections 5 and 7. Declining real interest rates and improved monitoring offer complementary readings of the same period, but both also operate by making collusion easier to sustain and thus reinforce, rather than undermine, the broader conclusion that stability favored coordination. The policy implication is a trade-off: stabilization policy is desirable as it compresses the demand cycle, but that same compression relaxes the constraint on tacit collusion, and it does so most in already concentrated markets—a competition cost that current mandates do not weigh against the stabilization benefit. Competition authorities should absorb it: cooperating with central banks, they should scrutinize mergers most closely where demand is stable and concentration is high, the conditions under which repeated

interaction is most likely to sustain elevated prices.

A Proofs

All results follow from the master relation $D(\bar{p}) = \sigma\chi(\delta, N)$ with $\chi(\delta, N) = (N-1)(1-\delta)/[N\delta - (N-1)]$. We collect the algebra here.

Reduction of the IC to Equation (4). Substituting $V^c(p) = (p-c)D(p)/[N(1-\delta)]$ into the peak-state constraint (3) and cancelling $(p-c)/N > 0$,

$$\delta \frac{D(p)}{1-\delta} \;\geq\; (N-1)\big(D(p)+\sigma\big).$$

Collecting the $D(p)$ terms,

$$D(p)\big[\delta-(1-\delta)(N-1)\big] \;\geq\; (1-\delta)(N-1)\sigma \iff D(p)\big[N\delta-(N-1)\big] \;\geq\; (1-\delta)(N-1)\sigma,$$

which is $D(p) \geq \sigma\chi(\delta, N)$. The bracket $N\delta - (N-1) > 0$ if and only if $\delta > (N-1)/N$, so the inequality is informative exactly on the feasible range.

Proof of Lemma 1. $\chi > 0$ requires $\delta > (N-1)/N$. Given this, $\sigma\chi > 0$, and a price $p > c$ with $D(p) \geq \sigma\chi$ exists if and only if $D(c) > \sigma\chi$ (since D is strictly decreasing and $D(p) < D(c)$ for $p > c$). The constraint binds at the highest price, so $\bar{p}$ solves $D(\bar{p}) = \sigma\chi$; strict monotonicity of D makes the root unique and yields $\bar{p} = D^{-1}(\sigma\chi)$. □

Sign of the χ-derivatives. Write the denominator $B(\delta, N) \equiv N\delta - (N-1) > 0$ on the feasible range.

$$\frac{\partial\chi}{\partial\delta} = (N-1)\frac{-B-(1-\delta)N}{B^2} = \frac{-(N-1)}{B^2} \;<\; 0,$$
$$\frac{\partial\chi}{\partial N} = (1-\delta)\frac{B-(N-1)(\delta-1)}{B^2} = (1-\delta)\frac{\delta}{B^2} \;>\; 0.$$

(The numerator of $\partial\chi/\partial\delta$ simplifies to -1 after expanding $-B-(1-\delta)N = -(N\delta - N + 1) - N + N\delta = -1$; that of $\partial\chi/\partial N$ simplifies to δ.) □

Proof of Proposition 1 and 2. Differentiating $\bar{p} = D^{-1}(\sigma\chi)$ and using $(D^{-1})' = 1/D' < 0$:

$$\frac{\partial\bar{p}}{\partial x} = \frac{1}{D'(\bar{p})}\frac{\partial(\sigma\chi)}{\partial x}, \qquad x \in \{\sigma, N, \delta\}.$$

With $\partial(\sigma\chi)/\partial\sigma = \chi > 0$ and $\partial\chi/\partial N > 0$ we get $\partial\bar{p}/\partial\sigma < 0$ and $\partial\bar{p}/\partial N < 0$; with $\partial\chi/\partial\delta < 0$ we get $\partial\bar{p}/\partial\delta > 0$. □

Proof of Proposition 3. Take logs of Equation (5): $\ln D(\bar{p}) = \ln \sigma + \ln \chi(\delta, N)$. Differentiating with respect to $\ln \sigma$ (holding δ, N fixed, so $\ln \chi$ is constant),

$$\frac{D'(\bar{p})\,\bar{p}}{D(\bar{p})} \cdot \frac{\partial \ln \bar{p}}{\partial \ln \sigma} = 1 \iff -\epsilon_D(\bar{p})\,\beta = 1 \iff \beta = -\frac{1}{\epsilon_D(\bar{p})}. \qquad \square$$

Proof of Proposition 4. Averaging the cell-level slope, which equals $-1/\eta$ on mass share q and 0 on share $1-q$, gives $\hat{\beta} = -q/\eta \in [-1/\eta, 0]$ for $q \in [0,1]$. Hence $\hat{\eta}_{\text{slope}} = -1/\hat{\beta} = \eta/q \geq \eta$, proving (i). For (ii), $\mu^{\star}_{\text{slope}} = 1/(1+\hat{\beta}) = 1/(1-q/\eta) = \eta/(\eta - q)$; this is increasing in q on $[0,1]$, so $\mu^{\star}_{\text{slope}} \leq \eta/(\eta-1) = \mu^{\star}(\eta)$ with equality iff $q = 1$. Part (iii) is immediate: $\Delta = \mu^{\star}_{\text{slope}} - \hat{\mu} \geq 0$ and $\mu^{\star}_{\text{slope}} \leq \mu^{\star}(\eta)$ give $\hat{\mu} \leq \mu^{\star}_{\text{slope}} \leq \mu^{\star}(\eta)$, strictly when $q < 1$. $\square$

B NAICS-to-SIC Matching Methodology

Matching firm-level Compustat data (coded in NAICS) to the state–sector GDP panel (coded in SIC) requires a three-step procedure combining an official crosswalk with two manual correspondence tables.

Step 1: Official crosswalk. We start from the 1987 SIC to 2002 NAICS crosswalk published by the Census Bureau, covering 1,180 NAICS-6 codes. The firm-level dataset contains 1,321 unique 6-digit NAICS codes, of which 890 are found directly in the crosswalk. The remaining 431 codes—representing firms in Compustat but absent from the official table—are handled in Step 2.

Step 2: Manual crosswalk for missing codes. Remaining unmatched codes were assigned manually using industry descriptions, with assignments checked against SIC and NAICS definitions. After this addition, all 1,321 NAICS codes are mapped to a SIC sector.

Step 3: Resolving non-unique matches. Even when a NAICS code is present, it may map to multiple 2-digit SIC codes. Of the 1,611 total NAICS-6 codes in the expanded crosswalk, 1,608 resolve unambiguously to one GDP sector description. For the remaining 3, a second manual table assigns the single best-fitting SIC sector. The 2,431 observations that remain without a match after all steps are dropped from the analysis (roughly 1.3% of raw observations).

C Markup Estimation Details

We estimate firm-level markups following **?** and **?**. The production function approach recovers the markup as the product of the output elasticity of a variable input and the inverse input cost share. Assuming Cobb–Douglas production functions at the 2-

digit NAICS sector level, the output elasticity θ_s^v is estimated via the control-function approach. The Hall formula then gives:

$$\mu_{it} = \theta_s^v \times \frac{P_{it}^y Q_{it}}{P_{it}^v V_{it}},$$

where $P^y Q$ is revenue (Compustat: `sale`) and $P^v V$ is variable input expenditure (Compustat: `cogs`). We use a Python implementation applied to the full Compustat panel.

Figure 7 plots the markup distribution before and after the deregulation, showing that after deregulation the markup distribution shifted to the right. Moreover, there seems to be no large difference in the impact of deregulation depending on whether the state–sector had a lower or higher volatility in the 1960s compared to the median state–sector.

D Descriptive Statistics

Table 10 reports summary statistics for the main analysis sample. The mean markup (sales-weighted) is 1.269, with a median of 1.267, indicating a roughly symmetric distribution around the mean. Moving average backward looking 3-year volatility is 0.094 with a standard deviation of 0.108, reflecting large cross-sectional heterogeneity. About 32% of observations fall in the post-deregulation period.

Table 10: Summary Statistics

Variable	Mean	SD	P10	P50	P90	N
Markup	1.269	0.240	0.998	1.267	1.529	27,221
Log Volatility	0.090	0.098	0.017	0.058	0.197	27,221
Baseline volatility	0.080	0.093	0.019	0.053	0.169	27,221
Baseline HHI	0.739	0.274	0.305	0.819	1.000	27,221
Post-deregulation indicator	0.318	0.466	0.000	0.000	1.000	27,221
Sales	7,054.914	20,752.267	51.131	1,204.988	16,252.427	27,221

Compustat coverage expands over the pre-restriction period—from 1,289 firms in 545 state–sectors in 1963 to 8,290 firms in 1,305 state–sectors by 1997 (Table 11)—mirroring the secular growth of the listed universe. We limit our analysis to the 1,111 state–sectors which were present in 1975–77.

Table 11: Sample Coverage Over Time

Year	Firms	Active state–sectors	Mean markup
1963	1,289	545	1.187
1964	1,379	567	1.194
1965	1,467	594	1.203
1966	1,563	612	1.201
1967	1,651	640	1.198
1968	1,854	685	1.203
1969	1,959	706	1.208
1970	2,026	720	1.203
1971	2,149	742	1.213
1972	2,263	765	1.205
1973	2,593	815	1.197
1974	3,471	977	1.171
1975	3,562	982	1.185
1976	3,631	996	1.190
1977	3,748	1,020	1.192
1978	3,873	1,042	1.195
1979	4,039	1,062	1.195
1980	4,343	1,090	1.180
1981	4,637	1,119	1.188
1982	5,186	1,157	1.214
1983	5,596	1,186	1.236
1984	5,869	1,214	1.250
1985	6,371	1,257	1.265
1986	6,525	1,265	1.294
1987	6,510	1,272	1.299
1988	6,351	1,254	1.303
1989	6,208	1,232	1.303
1990	6,206	1,216	1.306
1991	6,307	1,209	1.321
1992	6,633	1,243	1.323
1993	7,564	1,267	1.329
1994	7,847	1,284	1.325
1995	8,477	1,328	1.321
1996	8,551	1,336	1.324
1997	8,290	1,305	1.324

Notes: Firms counts distinct Compustat firms with non-missing sales-based markup; *Cells* the state–sectors, *Sectors* the SIC-2 sectors, and *States* the states covered in each year. *Mean markup* is the sales-weighted firm-level markup. Coverage expands as Compustat's listed universe grows.

Table 12: Single-firm state–sectors by year

Year	State-sectors	Single-firm	Share single
1966	425	159	0.374
1967	477	174	0.365
1968	496	169	0.341
1969	509	159	0.312
1970	518	155	0.299
1971	526	143	0.272
1972	535	131	0.245
1973	546	110	0.201
1974	589	28	0.048
1975	590	12	0.020
1976	590	7	0.012
1977	588	5	0.009
1978	590	4	0.007
1979	592	8	0.014
1980	590	8	0.014
1981	590	7	0.012
1982	589	3	0.005
1983	587	7	0.012
1984	584	12	0.021
1985	583	15	0.026
1986	587	21	0.036
1987	584	28	0.048
1988	581	34	0.059
1989	573	42	0.073
1990	574	52	0.091
1991	565	52	0.092
1992	571	47	0.082
1993	568	39	0.069
1994	569	43	0.076
1995	571	46	0.081
1996	565	42	0.074
1997	559	42	0.075

Notes: Each observation is a state–sector–year cell in the headline estimation sample. *Single-firm* counts cells with exactly one Compustat firm in that year, for which HHI = 1 by construction; *Share single* is that count divided by *State-sectors*. Such cells cannot sustain the tacit-collusion (Rotemberg–Saloner) mechanism and are excluded in the multi-firm robustness cut (Table 16 and Table 17).

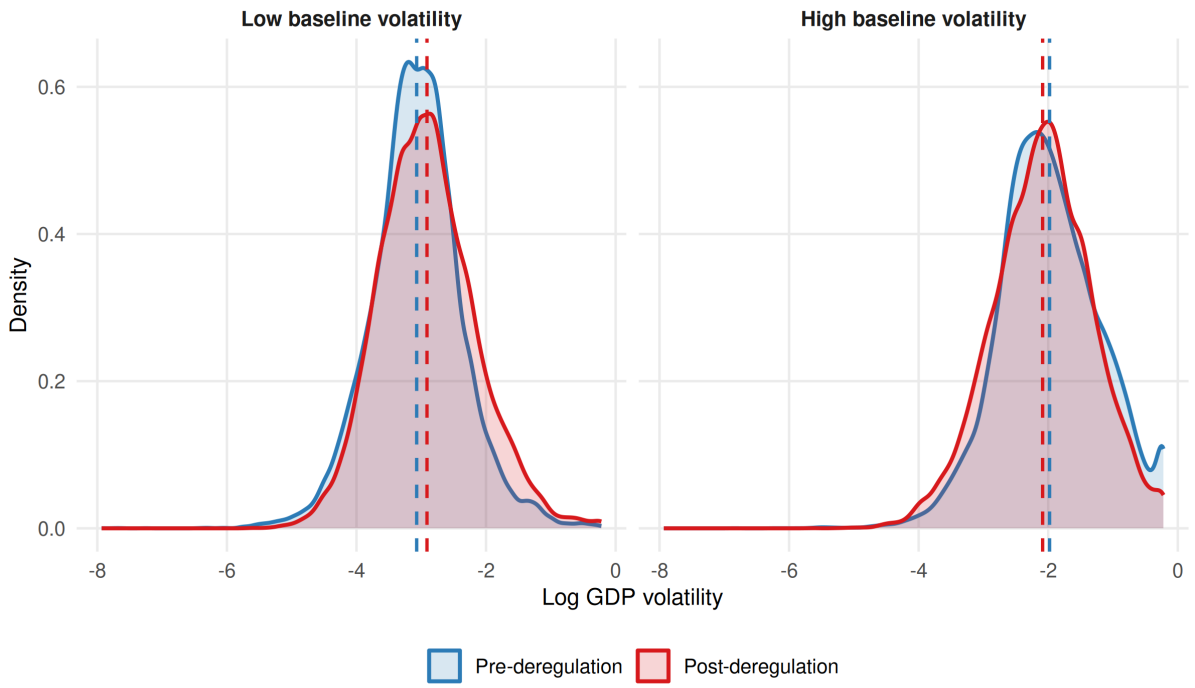


Figure 6: Distribution of state–sector volatility (GDP-weighted) before and after deregulation, for low and high baseline volatility.

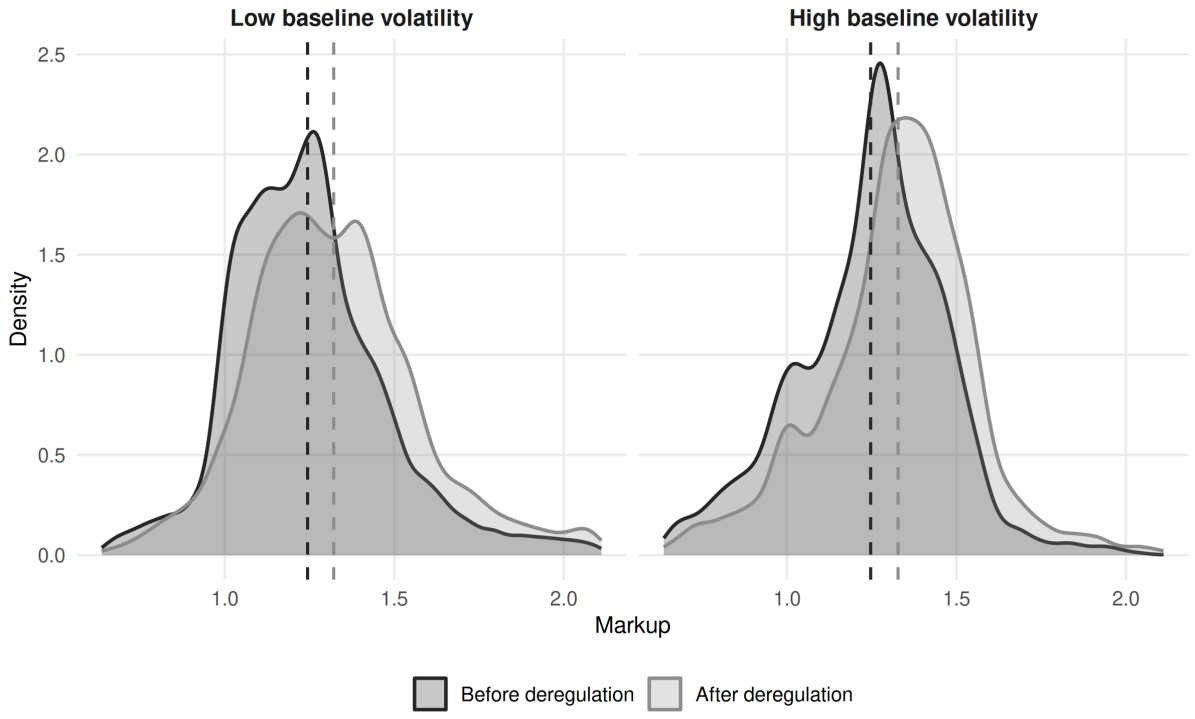


Figure 7: Distribution of state–sector markups (sales-weighted, μ_{sale}) before and after deregulation.

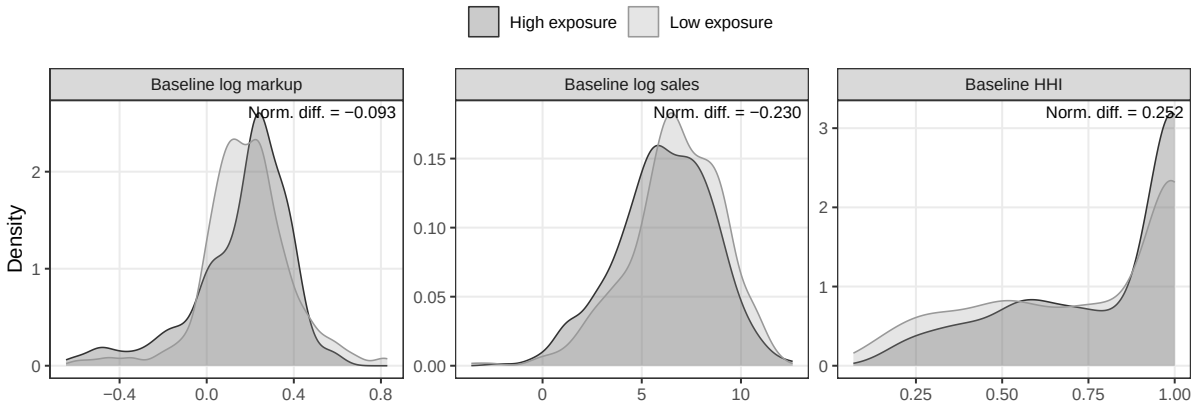


Figure 8: Predetermined state–sector characteristics by baseline-exposure group
Notes: Kernel densities of 1975–77 state–sector-level characteristics, by high/low exposure (median split). Overlapping distributions and small normalized differences support exposure exogeneity.

E Supplementary Robustness Checks

COGS weights. Table 13 re-weights firm markups within each state–sector by cost of goods sold rather than sales, holding the preferred concentration-interacted specification fixed. The two schemes differ in economically meaningful ways: because high-markup firms carry disproportionately larger revenue than cost weights, the sales-weighted markup loads more heavily on the upper tail of the distribution, whereas the cost-weighted markup is closer to the unweighted firm average and, as **?** argue, more directly tied to the aggregate distortion relevant for welfare [**?**]. Reassuringly, both the volatility effect and the HHI interaction are essentially invariant to sales versus COGs weighting. The volatility effect therefore reflects a shift in the markups within a market, not a reweighting toward a few high-markup firms.

Table 13: Robustness: Cost-of-Goods-Sold Weighted Markup

Dependent Variables:	ln_mu_sale	ln_mu_cogs
Specification	Sales markup	COGS markup
Model:	(1)	(2)
Variables		
Log Volatility	-0.241***	-0.242***
	(0.077)	(0.077)
Log Volatility × HHI	-0.595**	-0.598**
	(0.260)	(0.259)
Fit statistics		
Observations	27,221	27,221
Wald (1st stage), Log Volatility	19.230	19.230
Wald (1st stage), Log Volatility × HHI	15.511	15.511

Notes: Column (1) reproduces the headline sales-weighted specification; column (2) replaces the dependent variable with the log COGS-weighted state–sector markup (μ^{cogs}). SE clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Regional disaggregation. Table 14 re-estimates the concentration-interacted specification within each Census region. The level effect is negative in all four regions but it is only significant in the Northeast (-0.302^{***}). The HHI interaction is negative in three out of four regions and again it is only significant in the Northeast (-0.66^{***}). The loss of significance outside the Northeast is due to a loss of power of the first-stage, reflected in the low Wald statistic.

Table 14: By Census Macro-Region

Dependent Variable:	ln_mu_sale				
Specification	Pooled	Northeast	Midwest	South	West
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Log Volatility	-0.241***	-0.310***	-0.783	-0.050	-0.212
	(0.077)	(0.078)	(1.140)	(0.133)	(0.122)
Log Volatility × HHI	-0.595**	-0.660***	-2.082	0.353	-0.328
	(0.260)	(0.195)	(2.829)	(0.709)	(0.683)
Fit statistics					
Observations	27,221	5,992	7,160	9,288	4,781
Wald (1st stage), Log Volatility	19.230	180.48	9.3984	6.3242	7.0290
Wald (1st stage), Log Volatility × HHI	15.511	95.158	5.4021	1.1736	2.9899

Notes: Headline 2SLS specification estimated separately within each of the four Census Bureau regions. SE clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Dropping States. Table 15 confirms the result is not driven by any single state. Re-estimating the preferred concentration-interacted specification while dropping one state at a time, the volatility coefficient stays within $[-0.236, -0.157]$ around a baseline of -0.241, and the $\sigma \times$ HHI interaction within $[-0.597, -0.34]$ around -0.595. The mean jackknife values are essentially the same as the baseline and the tight dispersion—an order of magnitude smaller than the coefficients themselves—shows the headline estimate is not an artifact of any one state's markup or deregulation history.

Table 15: Robustness: State Jackknife

Statistic	Volatility	Volatility.x.HHI
Baseline coefficient	-0.241	-0.595
Jackknife min	-0.262	-0.694
Jackknife max	-0.181	-0.428
Jackknife mean	-0.241	-0.594
Jackknife SD	0.013	0.042

Notes: Leave-one-state-out jackknife of the preferred concentration-interacted IV (state–sector and year fixed effects). Each row summarizes the distribution of the coefficient across the 51 fits that drop one state. "Min"/"max" are the most extreme leave-one-out estimates; "mean" and "SD" summarize the full distribution.

Table 16 drops all state–sectors that had only one firm in the baseline years between 1975–77. The coefficients have the same sign, are significant and actually increase in absolute value.

Table 16: Robustness: sector exclusions, single-firm state–sectors dropped

Dependent Variable:	ln_mu_sale				
Specification	Baseline	No agric.	No mining	No RE	No finance
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Log Volatility	-0.272***	-0.272***	-0.260***	-0.267***	-0.268
	(0.065)	(0.065)	(0.071)	(0.063)	(0.221)
Log Volatility × HHI	-0.765***	-0.765***	-0.764***	-0.751***	-0.662
	(0.249)	(0.249)	(0.212)	(0.244)	(0.778)
Fit statistics					
Observations	17,961	17,961	17,435	17,599	16,667
Wald (1st stage), Log Volatility	18.070	18.070	25.812	20.239	6.0577
Wald (1st stage), Log Volatility × HHI	15.688	15.688	18.314	15.630	6.1235

Notes. The headline HHI-interacted 2SLS specification is re-estimated after dropping state–sectors whose average number of Compustat firms in the predetermined 1975–77 baseline window equals one. Columns sequentially exclude agriculture, mining, real estate, and finance (SIC 60–62). State×sector and year fixed effects are included throughout. Standard errors are clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 17: Robustness: disaggregating finance, single-firm state–sectors dropped

Dependent Variable:	ln_mu_sale					
Specification	Full sample	Baseline (multi-firm)	No finance	No bank	No credit	No sec.
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Log Volatility	-0.241***	-0.272***	-0.268	-0.287***	-0.206*	-0.300***
	(0.077)	(0.065)	(0.221)	(0.067)	(0.118)	(0.067)
Log Volatility × HHI	-0.595**	-0.765***	-0.662	-0.782***	-0.564	-0.758***
	(0.260)	(0.249)	(0.778)	(0.260)	(0.479)	(0.254)
Fit statistics						
Observations	27,221	17,961	16,667	17,216	17,698	17,675
Wald (1st stage), Log Volatility	19.230	18.070	6.0577	15.223	12.585	9.1850
Wald (1st stage), Log Volatility × HHI	15.511	15.688	6.1235	14.848	7.1874	15.448

Notes. As in Table 16, on the multi-firm sample. Columns sequentially exclude depository institutions (SIC 60), non-depository credit (SIC 61), security and commodity brokers (SIC 62), and narrowly-defined finance (SIC 60–62). *Full sample* retains single-firm state–sectors; *Baseline (multi-firm)* drops them with no sector exclusion, isolating the effect of the single-firm cut from the finance exclusions. State×sector and year fixed effects throughout; standard errors clustered by state. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.